

# **GEOGRAPHY**

# **PHYSICAL GEOGRAPHY**



## **GEOMORPHOLOGY**

### **MODULE - 30**

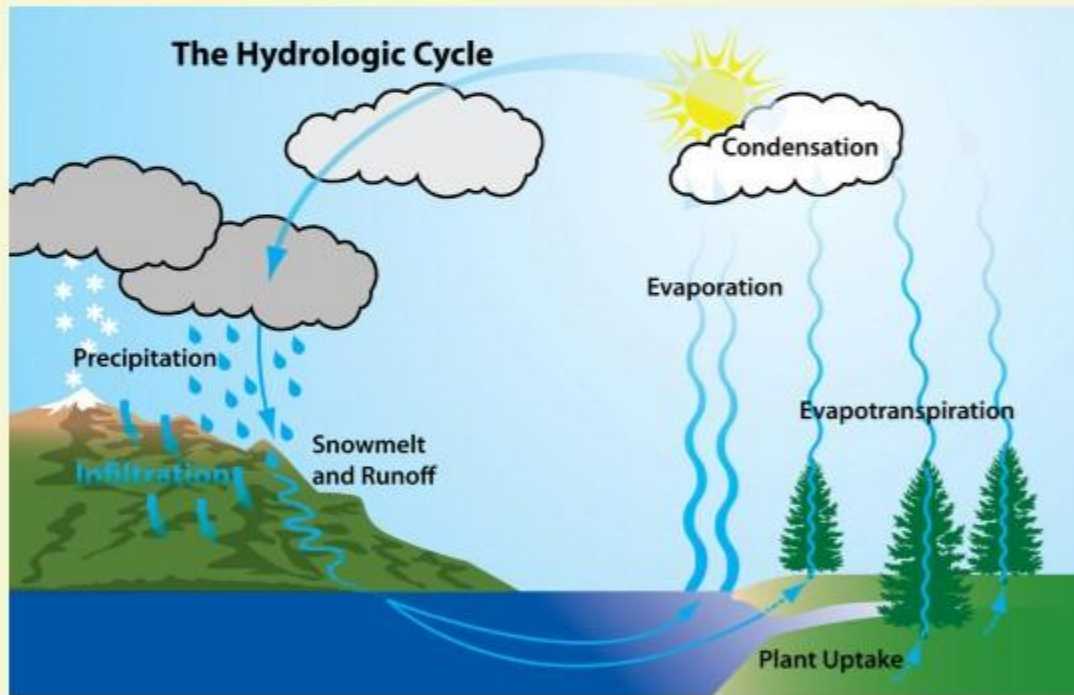
#### **GEOMORPHIC PROCESSES**

- POROSITY AND PERMEABILITY
- ARTESIAN WELLS



# MASS MOVEMENT

## HOW DOES GROUNDWATER AID IN MASS MOVEMENTS?



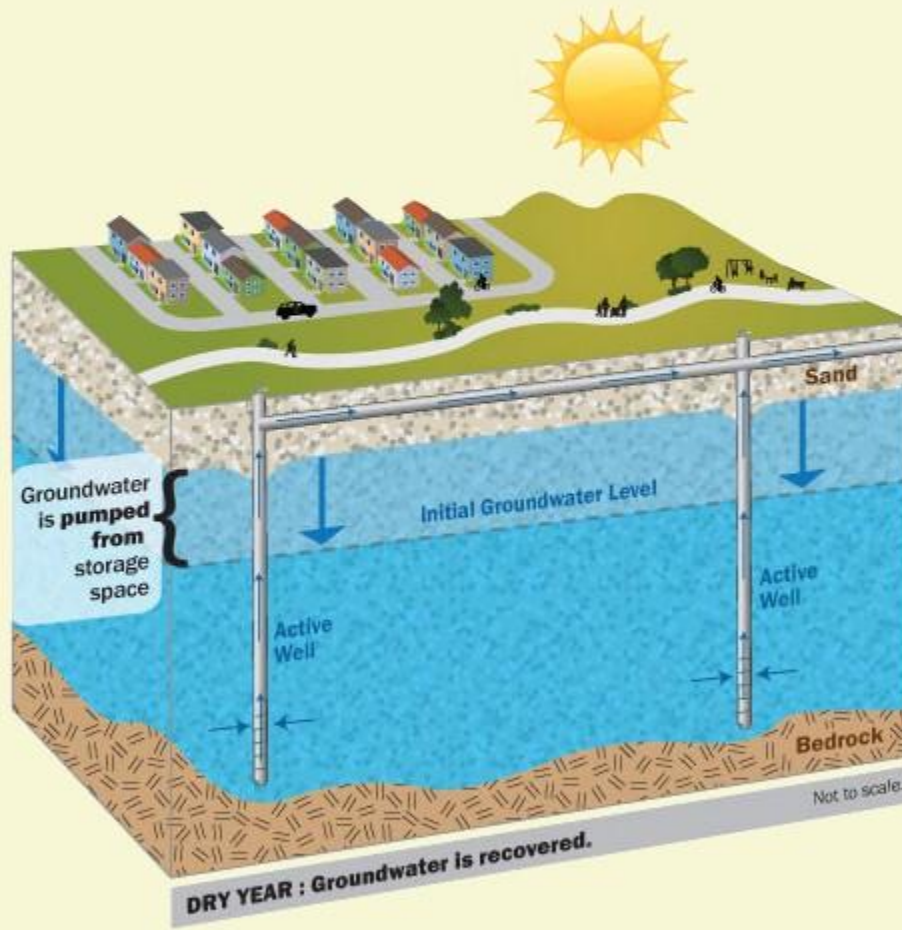
**1** The whole process of the circulation of water between the land, sea and atmosphere is known as the Hydrological cycle.

**2** When rain falls on the Earth it is distributed in various ways. Some is immediately evaporated and thus returns to the atmosphere as water vapour.

**3** Some is absorbed by plants and only gradually returned to the atmosphere by transpiration from the leaves of plants.

# MASS MOVEMENT

## HOW DOES GROUNDWATER AID IN MASS MOVEMENTS?



**3** Much of it flows directly off slopes to join streams and rivers, eventually reaching the seas and oceans. This is known as run-off.

**4** A considerable proportion of the water received from rain or snow, however, percolates downwards into the soil and rocks, filling up joints and pore-spaces and forming what is known as groundwater.

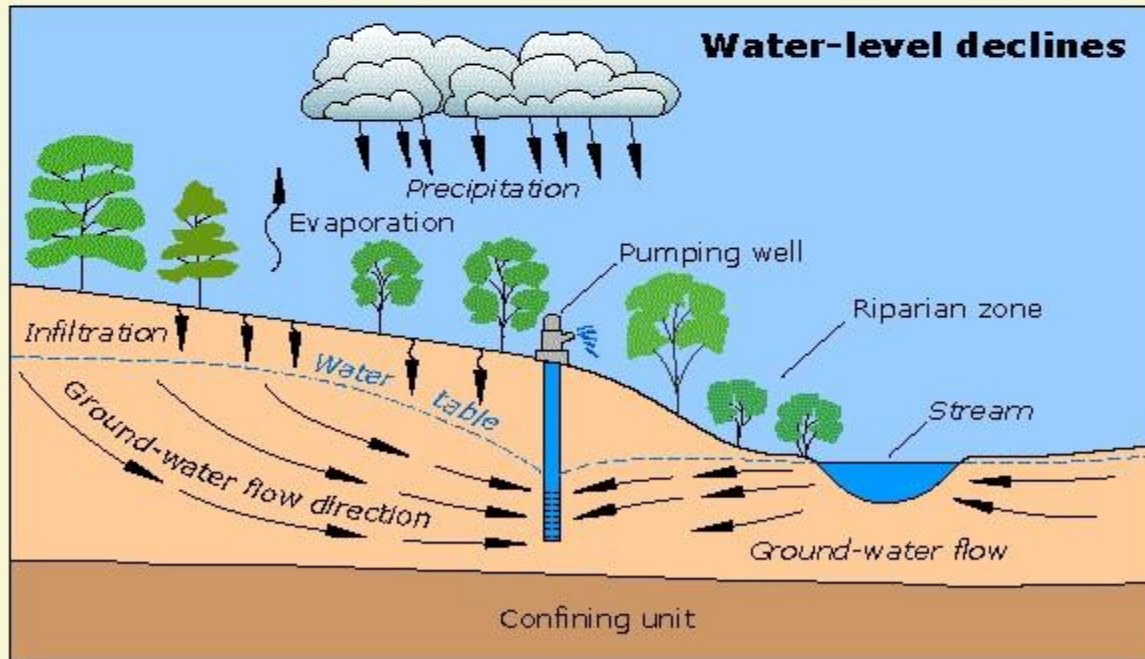
**5** Groundwater plays an important part in weathering and mass movement and is also important as a means of natural water storage.



# MASS MOVEMENT

## RELATIONSHIP BETWEEN GROUND WATER & CLIMATE

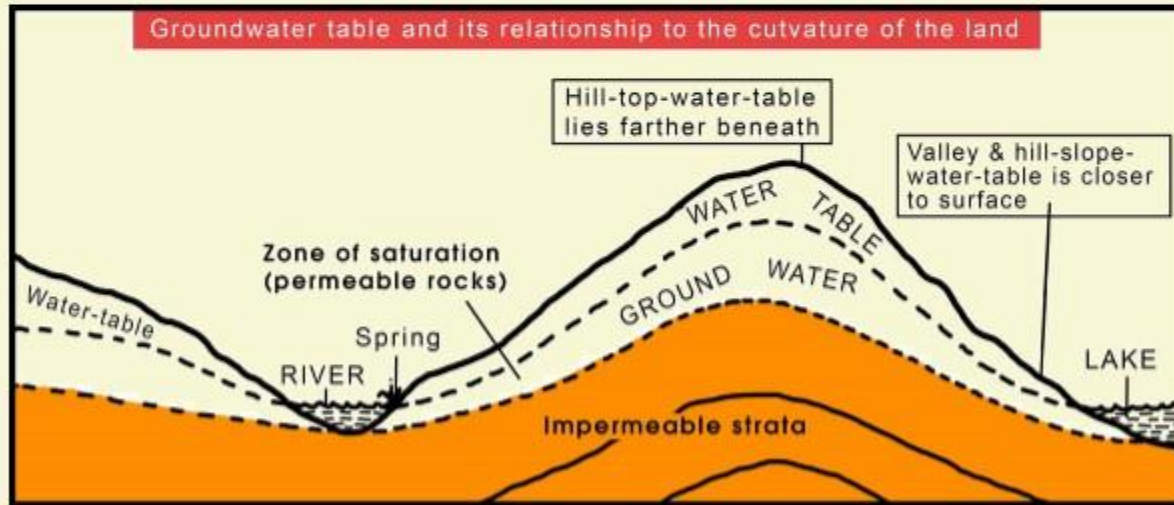
- 1 The amount of water available to form groundwater depends to **some extent on climate**.



- 2 In **dry climates** much precipitation may be quickly evaporated into the dry atmosphere and little moisture may percolate into the ground.
- 3 In **very humid conditions**, where the surface of the ground may already be moist, much water may be moved as run-off.
- 4 In **moderately humid areas**, water both runs off and sinks into the ground.

# MASS MOVEMENT

## RELATIONSHIP BETWEEN GROUND WATER & NATURE OF ROCKS



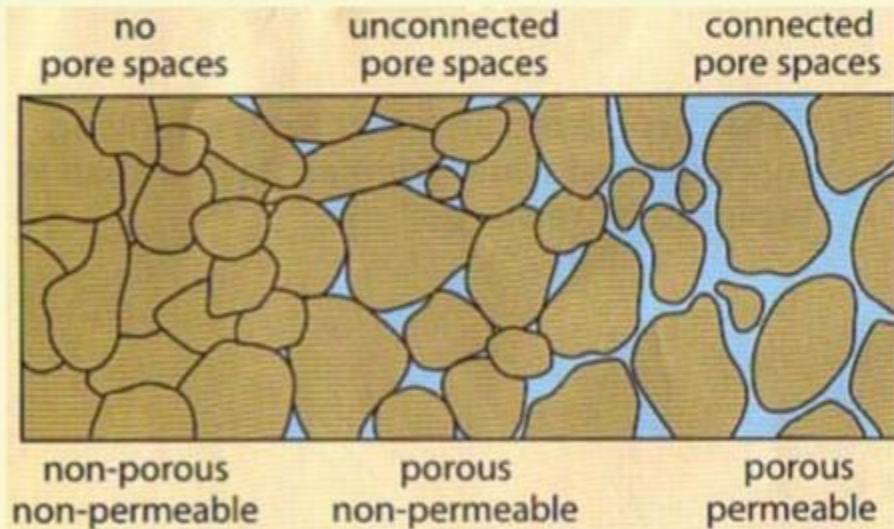
- 1 The proportion of the rainfall absorbed as groundwater may depend on the season of the year.
- 2 More important, however, is the nature of the rocks and how easily they absorb and retain water.
- 3 Various rocks and soils differ greatly in their porosity and permeability; the amount of groundwater present and the depth at which it lies are governed by these characteristics.



# MASS MOVEMENT

## RELATIONSHIP BETWEEN GROUND WATER & NATURE OF ROCKS

- 4** Porous rocks are those, like sandstone, which have many pore-spaces between the grains. Water is easily absorbed by such rocks and may be stored in the pore-spaces.



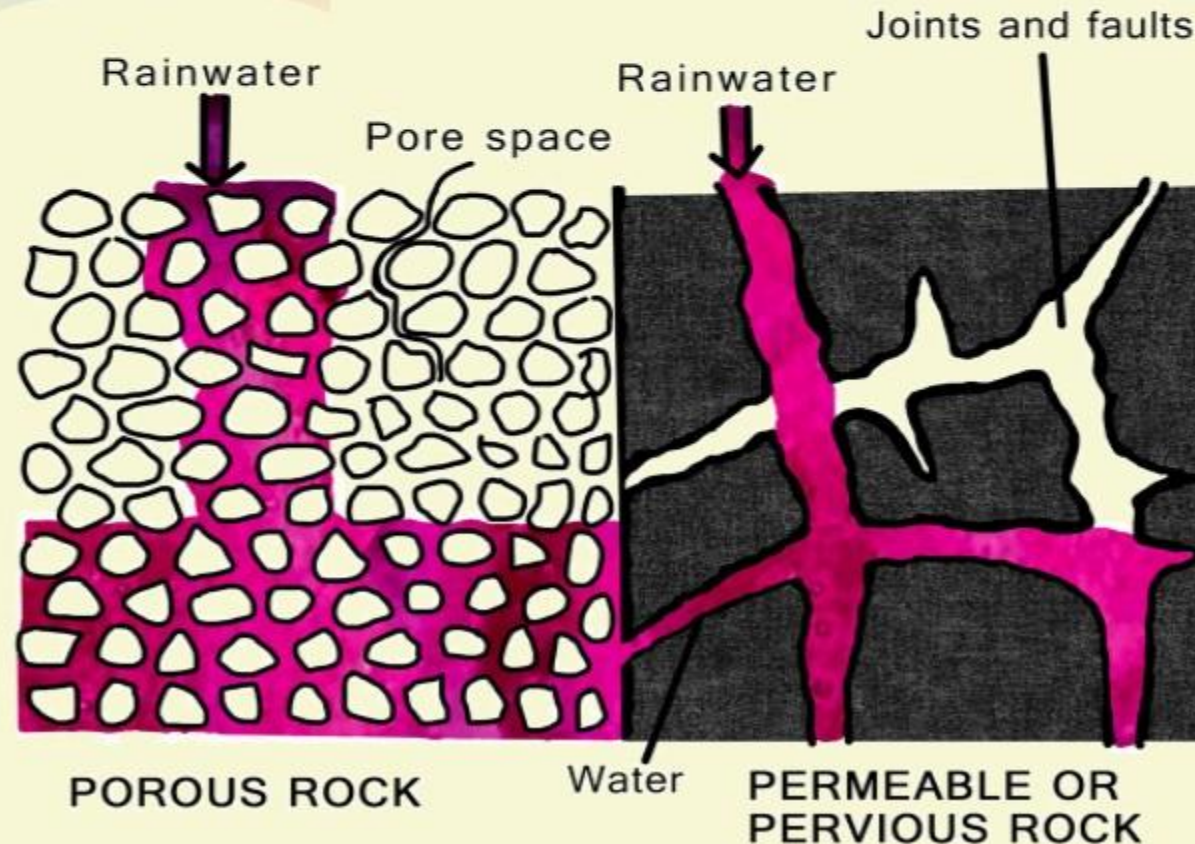
- 5** Permeable or pervious rocks are those which allow water to pass through them easily.
- 6** Thus most porous rocks are also permeable. However some rocks are porous but impermeable. Clay, for example, is highly porous since it is made up of innumerable very fine particles with pore-spaces between them. It thus absorbs a great deal of water.
- 7** However, the pore-spaces are so small that the water does not move easily through the rock, which is thus impermeable.



# MASS MOVEMENT

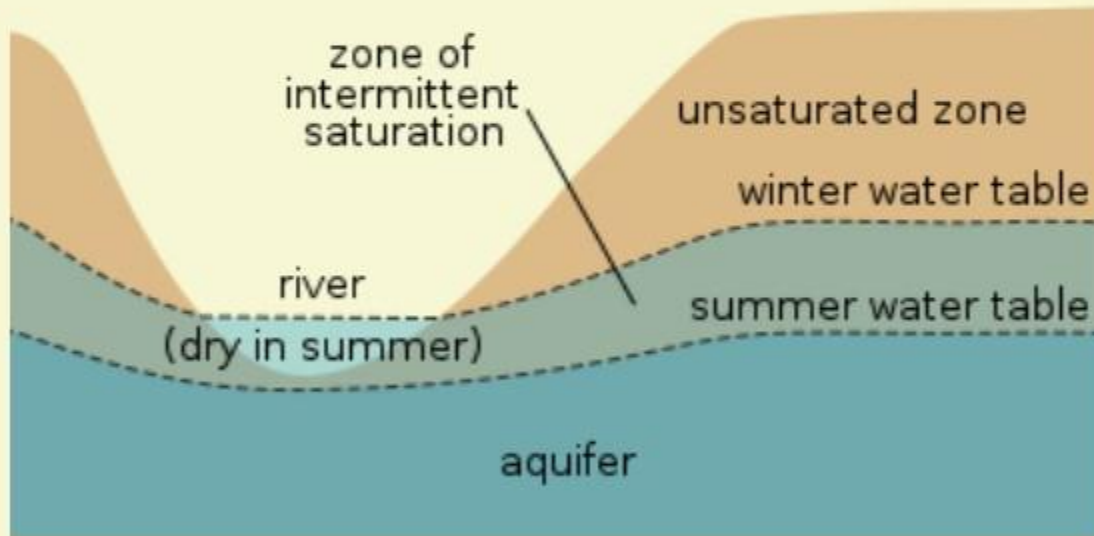
## HOW DOES GROUNDWATER AID IN MASS MOVEMENTS?

- 8 On the other hand, **granite** which is **crystalline rock** and consequently **non-porous** is often **pervious**. Its individual crystals absorb little or no water but the rock may have numerous **joints or cracks** through which the water can pass, **rendering it pervious or permeable**.



# MASS MOVEMENT

## RELATIONSHIP BETWEEN GROUNDWATER & WATER-TABLE

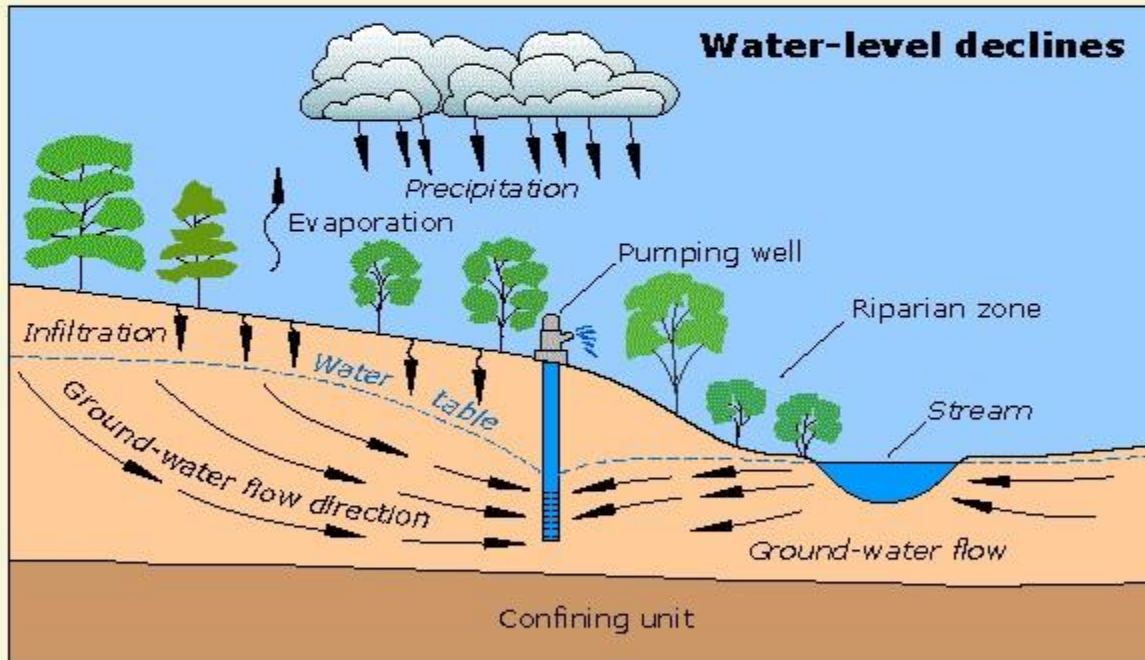


- 1 Water which seeps through the ground moves downward under the force of gravity until it reaches an impermeable layer of rock through which it cannot pass.
- 2 If there is no ready outlet for the groundwater in the form of spring, the water accumulates above the impermeable layer and saturates the rock.
- 3 The permeable rock in which the water is stored is known as aquifer. The surface of the saturated area is called the **Water-Table**.



# MASS MOVEMENT

## RELATIONSHIP BETWEEN GROUNDWATER & WATER-TABLE



**4** The depth of the water-table varies greatly according to relief and to the type of the rocks.

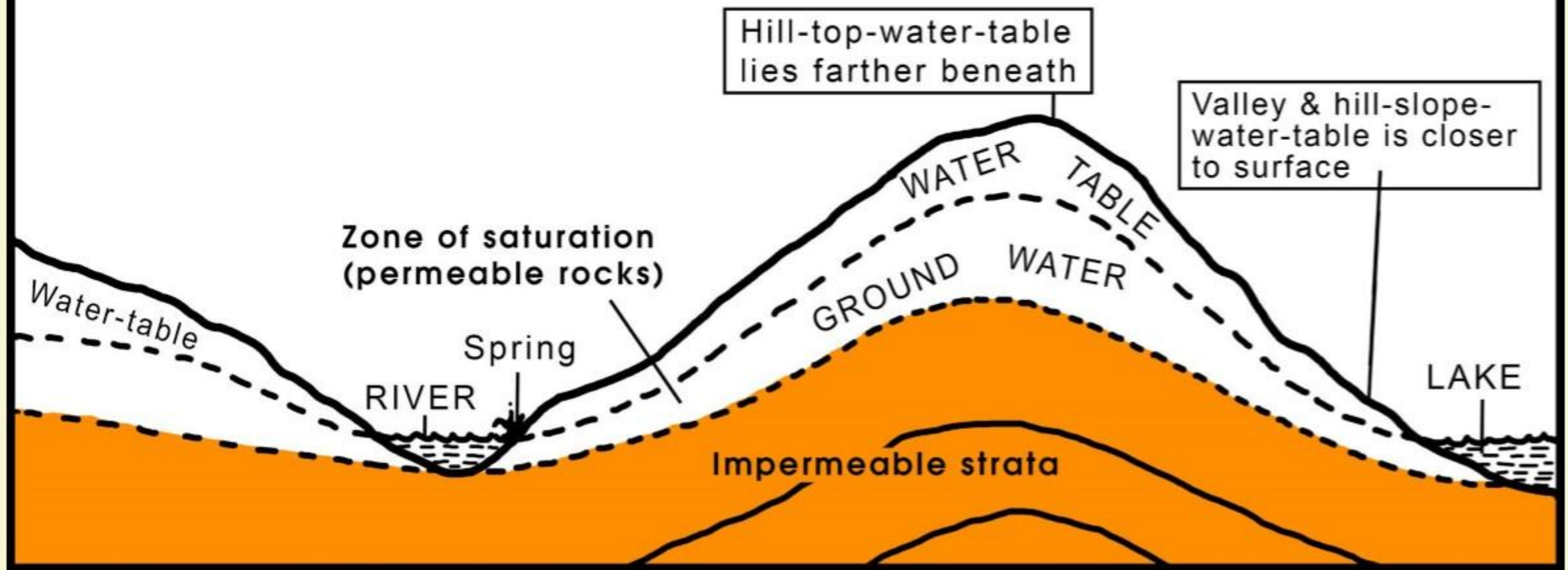
**5** The water-table is far below the surface of hill-tops but is close to the surface in valleys and the flat low-lying areas where it may cause waterlogging and swampy conditions.

**6** The depth of the water-table also varies greatly with the seasons. When plenty of rain is available to augment groundwater supplies the water-table may rise. But in dry periods, no new supplies are available, and the water-table is lowered as groundwater is lost through seepages and springs.

# MASS MOVEMENT

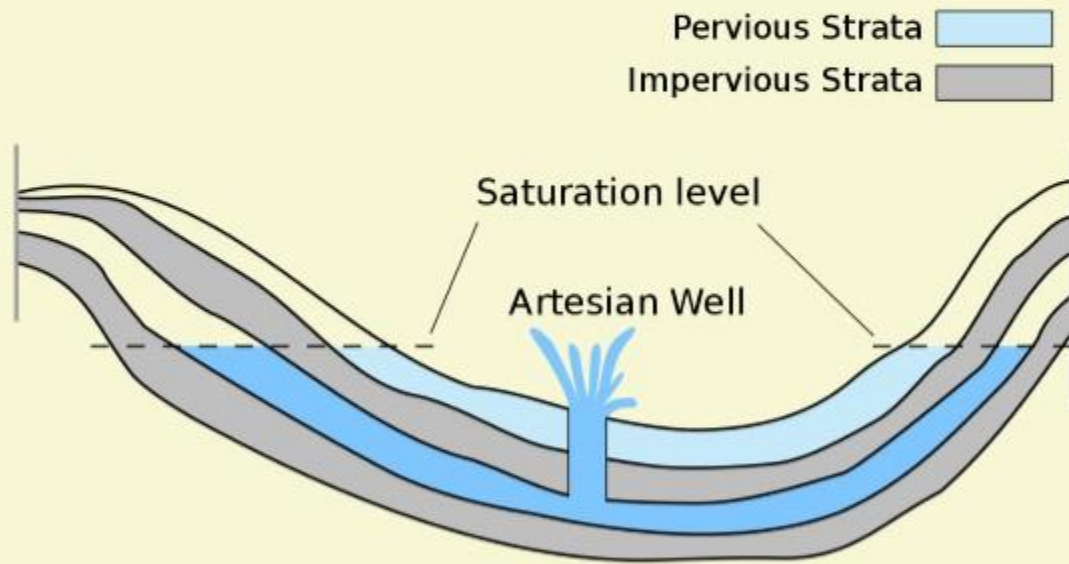
## HOW DOES GROUNDWATER AID IN MASS MOVEMENTS?

Groundwater table and its relationship to the curvature of the land



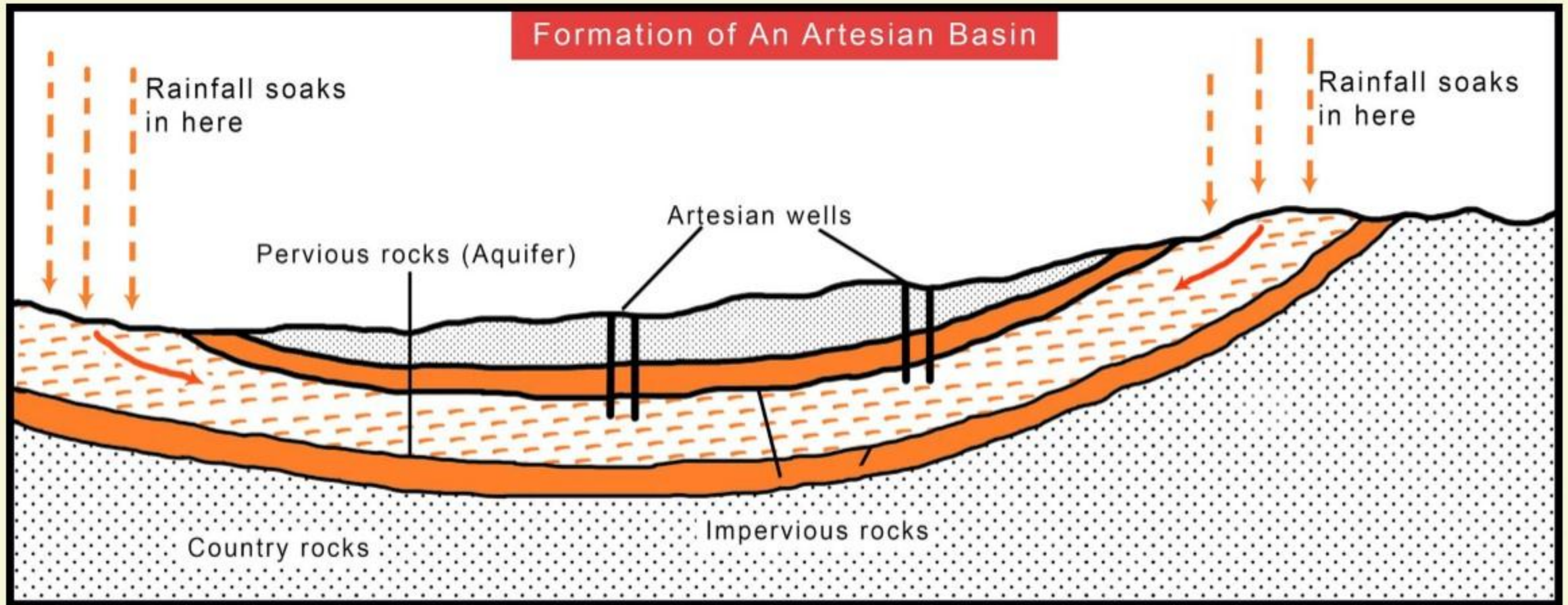


## INTERESTING FEATURE OF ARTESIAN WELLS...



- 1** A particularly important type of well is the Artesian well, which owing to the nature of its formation is quite distinctive.
- 2** Where rock layers have been down-folded into a basin shape, permeable strata such as chalk or limestone may be sandwiched between impermeable layers, such as clay.
- 3** The permeable rocks may only come to the surface at the edges of the basin, but water falling on them will gradually seep downwards by the force of gravity until it reaches the lowest part of the basin.

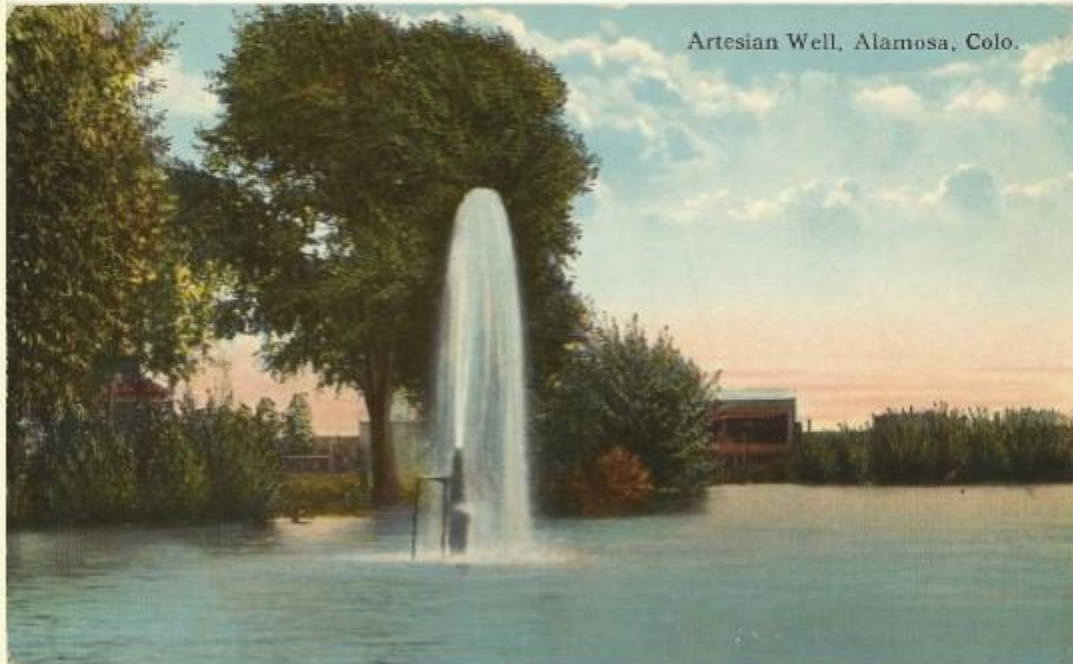
## INTERESTING FEATURE OF ARTESIAN WELLS...





## INTERESTING FEATURE OF ARTESIAN WELLS...

- 4** The impermeable layer below prevents the water from passing downwards while the impermeable layer on top prevents any possibility of the water escaping upwards.

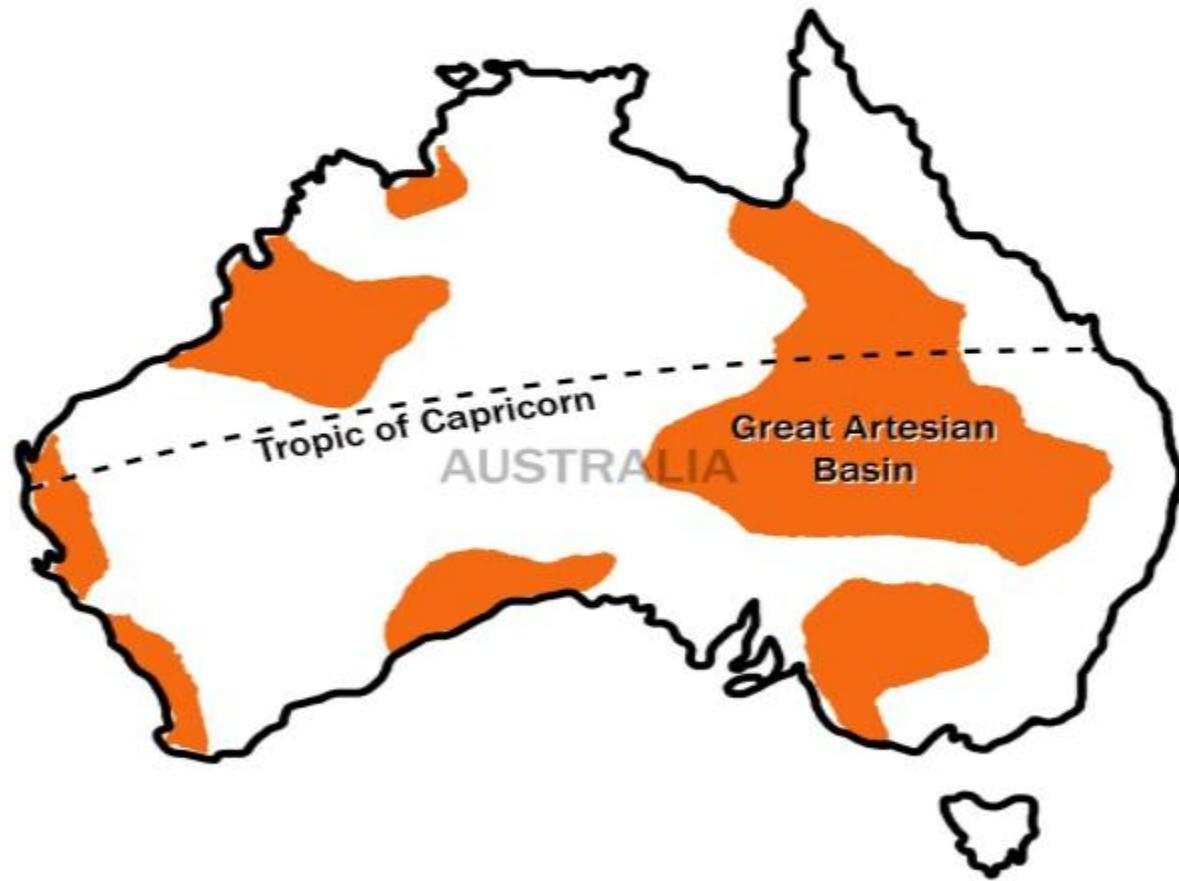


Artesian Well, Alamosa, Colo.

- 5** The aquifer is thus saturated to the brim of the basin. The water is thus trapped in the aquifer under great pressure.
- 6** When a well is bored, the pressure of the water downwards from all around the basin is sufficient to force the water up the bore-hole so that it gushes onto the surface like a fountain.
- 7** After a time the pressure decreases and it is necessary to pump up the water.

## INTERESTING FEATURE OF ARTESIAN WELLS...

The Distribution of Artesian Basins in Australia



### GREAT ARTESIAN BASIN

- 1 The Great Artesian Basin located in Australia is the largest and deepest artesian basin in the world, stretching over 17,00,000 square kilometres with measured temperatures ranging from 30–100 °C
- 2 The basin provides the only source of fresh water through much of inland Australia.



WE EXPRESS OUR  
SINCERE THANKS  
FOR VIEWING THIS VIDEO

Presented by



Learning Space

For Suggestions:

[suggestions@learningspace.in](mailto:suggestions@learningspace.in)

To Contact us:

[info@learningspace.in](mailto:info@learningspace.in)

**OUR TEAM**

G. V. Rao

K. Srikanth

D. Sunil Kumar

L. V. Krishna

D. Chaitanya

Amrita Naidu

M. Binukrishna

G. Ravi Babu

D. Joji

K. Victor Babu

Visit us at:

[www.learningspacedigital.com](http://www.learningspacedigital.com)

0 8 6 6 - 2 4 4 4 7 2

0 9 8 4 9 9 4 2 2 9 9